

SKY BREEZE ACADEMY

Group of Sky Breeze AC Solutions

Air Conditioning & HVAC Training Institute

Coimbatore, Tamilnadu

VOCATIONAL TRAINING PROGRAM

Diploma in Refrigeration & Air Conditioning (HVAC – Vocational)

Program Type:

1-Year Work-cum-Training Program

(TNSCVT – Government of Tamil Nadu)

Target Candidates:

10th Pass / Appeared

ITI Pass / Appeared

Diploma Pass / Appeared



PROGRAM OBJECTIVE

To develop competent HVAC technicians with strong practical skills in installation, servicing, troubleshooting, and maintenance of refrigeration and air conditioning systems, aligned with industry and TNSCVT standards.



SYLLABUS STRUCTURE (VOCATIONAL LEVEL)

MODULE 1: BASIC REFRIGERATION PRINCIPLES

- Introduction to refrigeration and air conditioning
 - Vapour compression refrigeration cycle
 - Refrigerants and their properties
 - Units of temperature, pressure, and heat
 - Safety practices in refrigeration work
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MODULE 2: AIR CONDITIONING SYSTEMS (DOMESTIC & LIGHT COMMERCIAL)

- Room Air Conditioners (High-wall split AC)
- Cassette air conditioners (1-way & 4-way)
- Ductable air conditioning units
- Package air conditioning units
- Application-based system selection

MODULE 3: HVAC COMPONENTS & FUNCTIONS

- Compressors used in RAC and light commercial AC
- Condensers and evaporators
- Expansion devices (capillary, TXV, EEV basics)
- Fans, motors, and blowers
- Filters and air flow components

MODULE 4: ELECTRICAL & CONTROL FUNDAMENTALS

- Electrical symbols and wiring diagrams
- Single-phase and three-phase motors
- Starting devices and protection components
- PCB-based control boards (basic understanding)
- Safety devices and interlocks

MODULE 5: INSTALLATION PRACTICES

- Tools and equipment used in HVAC work
- Copper tubing, flaring, bending, and brazing
- Insulation and drain piping
- Indoor and outdoor unit installation methods
- Site safety and workmanship standards

MODULE 6: SYSTEM COMMISSIONING

- Pressure testing and leak detection
- Evacuation and vacuuming procedures
- Refrigerant charging methods
- Performance checking and adjustments
- Start-up and shutdown procedures

MODULE 7: SERVICING & TROUBLESHOOTING

- Common faults in split and cassette ACs
- Electrical and refrigeration fault diagnosis
- Preventive maintenance procedures
- Breakdown servicing practices

- Customer handling and service reporting
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MODULE 8: INTRODUCTION TO COMMERCIAL HVAC SYSTEMS

- Basic awareness of VRF/VRV systems
 - Introduction to AHU and FCU systems
 - Overview of chiller systems
 - Role of technicians in commercial HVAC projects
 - Safety and coordination in large installations
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MODULE 9: WORKPLACE PRACTICES & PROFESSIONAL SKILLS

- Work discipline and safety compliance
 - Teamwork and communication on site
 - Documentation and job cards
 - Industry ethics and responsibilities
 - Introduction to career growth in HVAC field
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MODULE 10: ON-THE-JOB TRAINING (OJT)

- Real-time installation and service exposure
 - Assisting senior technicians on site
 - Gradual transition from observation to execution
 - Performance evaluation during training
 - Industry readiness assessment
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TRAINING MODE

- Classroom instruction (theory fundamentals)
 - Workshop practice
 - Continuous on-site training
 - Work-cum-learning model
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CERTIFICATION

On successful completion of training and examination,
TNSCVT – Government of Tamil Nadu
Vocational Certificate will be awarded.